i) Mean, Median, Mode, SD, SE and CV%

1. The arithmetic mean is calculated as:

A. ΣX / N

B. Middle value of the data

C. Most repeated value

D. SD / √N

Answer: A) ΣX / N

2. For an even number of observations, the median is:

A. The most frequent value

B. The first value

C. The average of the two middle values

D. The sum of all values

Answer: C) The average of the two middle values

3. The mode is the value that:

A. Lies in the middle

B. Occurs most frequently

C. Is the average

D. Is the largest

Answer: B) Occurs most frequently

4. A smaller standard deviation indicates that:

A. Data are widely scattered

B. The mean is zero

C. The sample is small

D. Data are more consistent

Answer: D) Data are more consistent

5. Standard Error (SE) of the mean is given by:

A. SD × √N

B. SD / √N

C. SD / Mean

D. Mean / SD

Answer: B) SD / √N

6. Coefficient of Variation (CV%) is:

A. (SD / Mean) × 100

B. (Mean / SD) × 100

C. SD / √N

D. ΣX / N

Answer: A) (SD / Mean) × 100

7. For the data 10, 20, 20, 30, 40, the median is:

A. 24

B. 30

C. 20

D. 40

Answer: C) 20

8. If SD = 4 and N = 16, the standard error is:

A. 0.25

B. 1

C. 4

D. 64

Answer: B) 1

ii) Test of Significance and Control Charts

9. The null hypothesis (H_0) assumes that:

A. A significant difference exists

B. The sample is biased

C. The process is out of control

D. There is no significant difference

Answer: D) There is no significant difference

10. The level of significance is the probability of:

A. Rejecting H_0 when it is actually true

B. Accepting H_0 when it is false

C. Selecting a random sample

D. Finding the mean

Answer: A) Rejecting H_0 when it is actually true

11. According to the basic decision rule, $p \leq 0.05$ is regarded as:

A. Not significant

B. Invalid

C. Significant

D. Undefined

Answer: C) Significant

12. The t-test is used mainly to compare:

A. Variances

B. Means, especially for small samples

C. Categorical data

D. Defect counts

Answer: B) Means, especially for small samples

13. The chi-square test (χ^2) is used for:

A. Comparing variances

B. Comparing means of small samples

C. Drawing control limits

D. Categorical data/association

Answer: D) Categorical data/association

14. The F-test is used to compare:

A. Variances

B. Means

C. Proportions defective

D. Moisture regain values

Answer: A) Variances

15. The general form of the control limits is:

A. CL ± SD/√N

B. CL ± 1σ

C. UCL = CL + 3σ and LCL = CL − 3σ

D. UCL = CL × 3

Answer: C) UCL = CL + 3σ and LCL = CL − 3σ

16. The R chart is used to monitor:

A. Process average

B. Range/variation

C. Fraction defective

D. Defects per unit

Answer: B) Range/variation

17. The np chart is used for:

A. Number of defective units

B. Number of defects

C. Average

D. Range

Answer: A) Number of defective units

18. The c chart is used for:

A. Fraction defective

B. Mean

C. Defects per unit

D. Number of defects (count)

Answer: D) Number of defects (count)

19. The p chart is used for:

A. Mean

B. Number of defects

C. Fraction/proportion defective

D. Range

Answer: C) Fraction/proportion defective

20. The u chart is used for:

A. Number of defective units

B. Defects per unit

C. Process average

D. Range

Answer: B) Defects per unit

iii) Sampling Techniques

21. Which of the following is NOT an objective of sampling?

A. Increase testing cost

B. Reduce testing time

C. Obtain representative information

D. Make decisions about the entire population

Answer: A) Increase testing cost

22. In simple random sampling:

A. Units are taken at a fixed interval

B. The population is divided into strata

C. Every unit has an equal chance of selection

D. Units are chosen for convenience

Answer: C) Every unit has an equal chance of selection

23. Systematic sampling involves selecting units:

A. Randomly from clusters

B. Based on accessibility

C. From each stratum

D. At a fixed interval

Answer: D) At a fixed interval

24. In stratified sampling:

A. Whole clusters are sampled

B. The population is divided into groups/strata and samples are taken from each

C. Samples are selected for ease

D. Every nth unit is chosen

Answer: B) The population is divided into groups/strata and samples are taken from each

25. In cluster sampling:

A. The population is divided into clusters and selected clusters are sampled

B. Every unit has an equal chance

C. Units are taken at fixed intervals

D. Samples are drawn from every stratum

Answer: A) The population is divided into clusters and selected clusters are sampled

26. Convenience sampling selects samples based on:

A. Equal probability

B. Fixed intervals

C. Ease or accessibility

D. Strata proportions

Answer: C) Ease or accessibility

iv) Humidity Control

27. The commonly specified standard atmosphere for general textile testing is:

A. 27 ± 2°C and 65 ± 5% RH

B. 20 ± 2°C and 65 ± 2% RH

C. 20 ± 2°C and 50 ± 2% RH

D. 30 ± 2°C and 85 ± 5% RH

Answer: B) 20 ± 2°C and 65 ± 2% RH

28. The commonly specified tropical standard atmosphere is:

A. 20 ± 2°C and 65 ± 2% RH

B. 20 ± 2°C and 50 ± 5% RH

C. 25 ± 2°C and 60 ± 2% RH

D. 27 ± 2°C and 65 ± 5% RH

Answer: D) 27 ± 2°C and 65 ± 5% RH

29. Relative humidity is the ratio of:

A. Actual water vapour present to the saturation water vapour capacity at the same temperature, as a percentage

B. Dry-bulb temperature to wet-bulb temperature

C. Moisture regain to fibre weight

D. Air pressure to vapour pressure

Answer: A) Actual water vapour present to the saturation water vapour capacity at the same temperature, as a percentage

30. Which instrument is used to measure humidity?

A. Lea tester

B. Trash analyser

C. Psychrometer

D. Micronaire

Answer: C) Psychrometer

31. Atmospheric conditioning is important because moisture affects:

A. Only fibre strength

B. Only weight

C. Only dimensions

D. Strength, elongation, weight, moisture regain, dimensions and yarn/fabric behaviour

Answer: D) Strength, elongation, weight, moisture regain, dimensions and yarn/fabric behaviour

v) Measurement of Fibre Properties

32. Trash in cotton refers to:

A. Degree of fibre development

B. Fibre thickness

C. Fibre length

D. Foreign matter/non-lint material

Answer: D) Foreign matter/non-lint material

33. Fibre maturity is particularly important in:

A. Cotton

B. Polyester

C. Nylon

D. Glass fibre

Answer: A) Cotton

34. Fibre fineness affects:

A. Only trash content

B. Only fibre length

C. Number of fibres in yarn cross-section, yarn appearance and handle

D. Only crease resistance

Answer: C) Number of fibres in yarn cross-section, yarn appearance and handle

35. Fibre length affects:

A. Only dyeing

B. Yarn strength, spinning performance and yarn evenness

C. Only GSM

D. Only colour fastness

Answer: B) Yarn strength, spinning performance and yarn evenness

36. The memory sequence L–S–F–M–T stands for:

A. Length, Softness, Finish, Moisture, Twist

B. Lustre, Strength, Fineness, Maturity, Twist

C. Length, Stiffness, Fabric, Maturity, Trash

D. Length, Strength, Fineness, Maturity, Trash

Answer: D) Length, Strength, Fineness, Maturity, Trash

37. Fibre strength (breaking force/tenacity) is measured using a:

A. Fibre tensile tester

B. Trash analyser

C. Psychrometer

D. Thermo-hygrometer

Answer: A) Fibre tensile tester

vi) Yarn Count, Twist and Strength

38. In direct systems such as Tex and Denier, a higher number means:

A. Finer yarn

B. Coarser yarn

C. Stronger yarn

D. More twist

Answer: B) Coarser yarn

39. In indirect systems such as Ne and Nm, a higher number means:

A. Coarser yarn

B. Heavier yarn

C. Finer yarn

D. Higher GSM

Answer: C) Finer yarn

40. Which of the following are direct yarn count systems?

A. Tex and Denier

B. Ne and Nm

C. CSP and RKM

D. TPI and TPM

Answer: A) Tex and Denier

41. Which of the following are indirect yarn count systems?

A. Tex and Denier

B. TPI and TPM

C. CSP and RKM

D. English cotton count (Ne) and metric count (Nm)

Answer: D) English cotton count (Ne) and metric count (Nm)

42. Twist is commonly expressed in:

A. Tex

B. Denier

C. Turns per inch (TPI) or turns per metre (TPM)

D. Grams per square metre

Answer: C) Turns per inch (TPI) or turns per metre (TPM)

43. Count Strength Product (CSP) for cotton yarn is:

A. Count + Lea breaking strength

B. Yarn count × Lea breaking strength

C. Count / Lea breaking strength

D. Lea strength − Count

Answer: B) Yarn count × Lea breaking strength

44. RKM (under its conventional definition) represents:

A. Yarn tenacity – the length in km whose breaking load equals its own mass in kg

B. Yarn twist

C. Yarn evenness

D. Yarn hairiness

Answer: A) Yarn tenacity – the length in km whose breaking load equals its own mass in kg

45. Elongation of a yarn is:

A. Its fineness

B. Its twist level

C. Its strength per unit count

D. The increase in length at break, expressed as a percentage

Answer: D) The increase in length at break, expressed as a percentage

46. A yarn of original length 250 mm extends by 5 mm at break. Its elongation is:

A. 0.5%

B. 5%

C. 2%

D. 20%

Answer: C) 2%

vii) Evenness, Imperfections and Hairiness

47. Common yarn imperfections include:

A. Twist and count

B. Thin places, thick places and neps

C. GSM and drape

D. Crease and pilling

Answer: B) Thin places, thick places and neps

48. Yarn hairiness refers to:

A. Fibres protruding from the main body of the yarn

B. Variation in twist

C. Variation in count

D. Yarn breaking load

Answer: A) Fibres protruding from the main body of the yarn

49. Uneven yarn can cause:

A. Only increased strength

B. Only improved handle

C. Only better dyeing

D. Barré, streaks, uneven dyeing and fabric appearance problems

Answer: D) Barré, streaks, uneven dyeing and fabric appearance problems

50. Excessive hairiness may cause:

A. Better evenness

B. Lower friction

C. Pilling, fuzziness, higher friction and processing difficulties

D. Higher CSP

Answer: C) Pilling, fuzziness, higher friction and processing difficulties

viii) Fabric Strength, Stiffness, Handle, Drape, Thickness and GSM

51. Fabric stiffness is the resistance to:

A. Bending

B. Breaking

C. Rubbing

D. Shrinking

Answer: A) Bending

52. Drape is the ability of a fabric to:

A. Resist abrasion

B. Resist creasing

C. Allow air to pass

D. Hang in folds under its own weight

Answer: D) Hang in folds under its own weight

53. A fabric piece of area 0.02 m^2 weighs 3 g. Its GSM is:

A. 15

B. 60

C. 150

D. 1500

Answer: C) 150

54. Common fabric strength tests include:

A. Drape and handle tests

B. Strip test, grab test and tensile test

C. Rubbing and light tests

D. Count and twist tests

Answer: B) Strip test, grab test and tensile test

ix) Crease, Abrasion, Pilling, Permeability and Dimensional Stability

55. Pilling occurs due to:

A. Fibre migration, entanglement and abrasion

B. Excess moisture only

C. Fabric thickness

D. Light exposure

Answer: A) Fibre migration, entanglement and abrasion

56. Air permeability is important for:

A. Only carpets

B. Only upholstery

C. Only geotextiles

D. Sportswear, filters, protective clothing and breathable fabrics

Answer: D) Sportswear, filters, protective clothing and breathable fabrics

57. Dimensional change of fabric may be:

A. Pilling or fading

B. Shrinkage or growth

C. Staining or bleeding

D. Creasing or drape

Answer: B) Shrinkage or growth

x) Colour Fastness

58. In fastness to washing, the observations made are:

A. Dry and wet colour transfer

B. Fading due to light

C. Fabric thickness

D. Colour change and staining of adjacent fabric

Answer: D) Colour change and staining of adjacent fabric

59. Fastness to rubbing is tested under which two conditions?

A. Hot and cold

B. Acid and alkali

C. Dry and wet

D. Light and dark

Answer: C) Dry and wet

60. Poor light fastness results in:

A. Fading

B. Colour transfer

C. Shrinkage

D. Pilling

Answer: A) Fading